

Stage 1: Input & Residual Extraction

- Classical Baseline Law: $f_{\text{base}}(X)$
- Observational Data: (X, y_{obs})
- Target Residual: $\delta = y_{\text{obs}} - f_{\text{base}}$

Stage 2: Candidate Proposer

- Stochastic Grammar Template Sampling
- LLM Vocabulary & Domain Hints
- Residual Feature Prior Bias

Stage 3: Cascaded Physics Gates

1. AST Complexity Gate: Depth ≤ 7 , Tokens ≤ 20
2. Dimensional Guardrail: Buckingham- Π Units
3. Asymptotic Limit Gate: $\lim_{X \rightarrow X_0} \Delta = 0$

Stage 4: Fitting & Occam's Selection

- JAX JIT-Traced L-BFGS-B Fitting
- Schwarz Bayesian Information Criterion (BIC)
- Validated Discovery: $\Delta(X; \theta)$

Feedback (if stuck)

